DESIGN-TIME QUANTUM / RUNTIME CLASSICAL

Quantum-Assisted Assurance for Autonomous Aviation

A research framework for training and verifying classical autonomous aircraft by using quantum-assisted exploration to expose rare decision-chain failures before deployment.

QAA maps unsafe autonomy behaviors during design, then distills those findings into classical runtime systems.

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THE CHALLENGE

Assurance is becoming the bottleneck

Formal verification is strongest when system behavior can be fully modeled. Autonomous aviation challenges that assumption because distributed, learning-based aircraft systems generate long chains of interdependent decisions across large state and action spaces.

Distributed behavior
Multiple subsystems interact in complex, emergent ways.

Decision-chain failures
Safety events arise from sequences, not isolated actions.

Non-static learning
Adaptive systems are difficult to model with static verification.

Exponential spaces
Classical simulation struggles to exhaustively cover rare states.

Decision-space growth

coverage gap widens as autonomy complexity increases

CORE IDEA

How QAA works

01 Explore large decision spaces
Quantum-assisted methods search autonomy scenarios that are hard to cover classically.

02 Identify rare unsafe behaviors
Low-probability, high-risk regions are surfaced before runtime deployment.

03 Feed classical AI training
Insights improve deterministic autonomous behavior while preserving certification compatibility.

SYSTEM DESIGN

Architecture

01 Quantum-Assisted Design Layer
Scenario exploration / policy stress-testing / off-board only

02 Classical Autonomous Execution
Deterministic AI / safety monitors / certifiable runtime behavior

03 Assurance and Verification Layer
Decision summaries / explainability / audit support

PROCESS

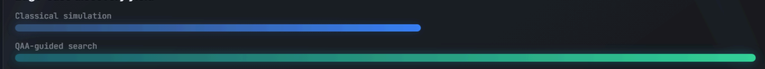
Methodology

PHASE 01
Classical Baseline
Establish decision policy, safety metrics, and autonomy task.

PHASE 02
Quantum-Assisted Exploration
Explore large decision spaces and surface rare unsafe behaviors.

PHASE 03
Hybrid Training
Integrate quantum-derived insights into robust classical AI.

Edge-case discovery yield



Training signal path

Scenario — Stress — Trace — Distill

SCOPE CLARIFICATION

What this research is not

Not proposed
- Quantum hardware onboard aircraft
- Quantum flight control
- Non-deterministic runtime behavior

Proposed
- Quantum used off-board
- Training, testing, and verification support
- Fully classical runtime AI

OPEN QUESTIONS

Research questions

RQ-01
Can quantum-assisted exploration expose unsafe autonomy behaviors faster than classical simulation?

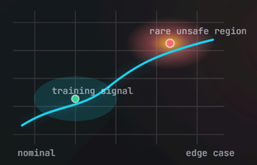
RQ-02
How can quantum-generated insights be distilled into classical, certifiable AI?

RQ-03
What does decision-level assurance mean for autonomous aviation?

RQ-04
How can verification scale without affecting real-time operation?

ASSURANCE VIEW

Risk regions



RESEARCH EXPANSION

QTRD: Quantum Tree-Based Rare-Event Distillation

A forward-looking extension uses a tree-based teacher model to identify rare, high-risk decision regions, quantum optimization to search those regions efficiently, and student AI distillation to improve deployable classical models.

Data to Teacher Forest — Quantum Rare Region Search — Edge Case Generation — Student AI Distillation

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